

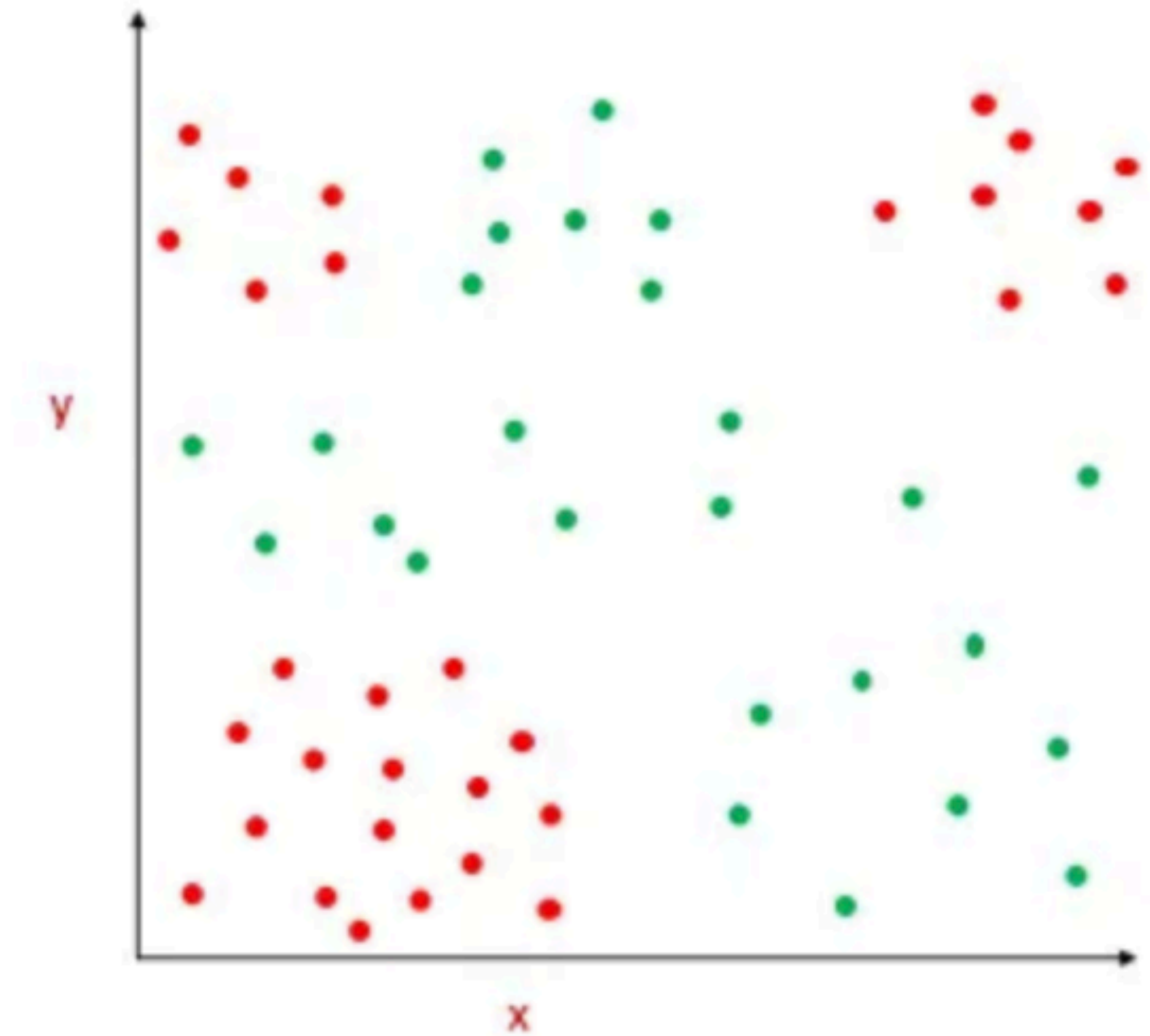
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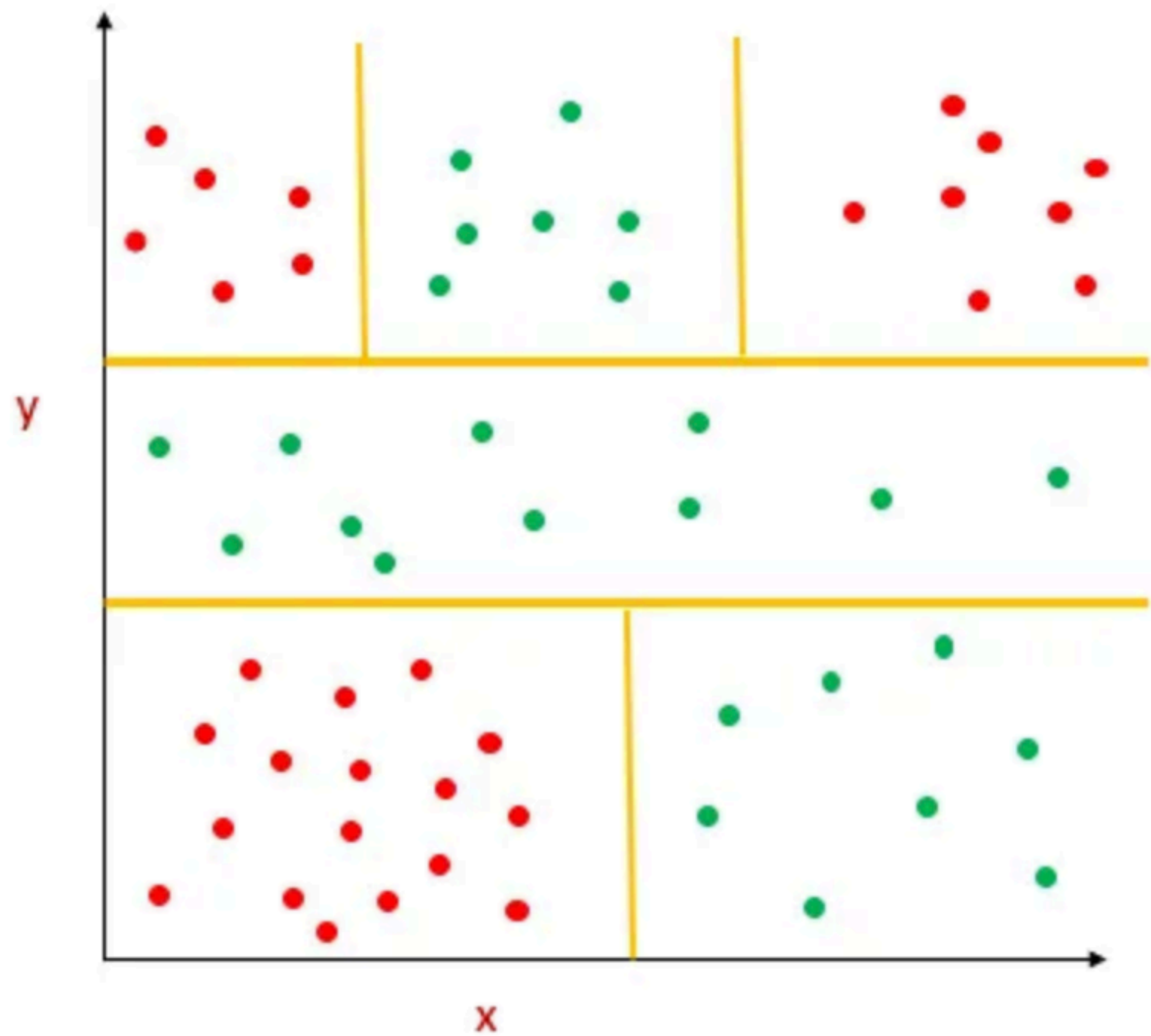
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tree-based methods

goal





- stratify predictor space into regions
- rules summarised as tree

- decision trees
- combination of models into better ones

decision trees

- can be applied to both classification and regression

types of nodes

- root: topmost node
- internal: nodes that split further
- leaf: nodes that do not split

how to stratify?

- based on the feature at each node
 - continuous variable
 - threshold
 - categorical variable
 - value

- example
 - $\text{years} < 4.5$
 - $\text{gender} = \text{male}$

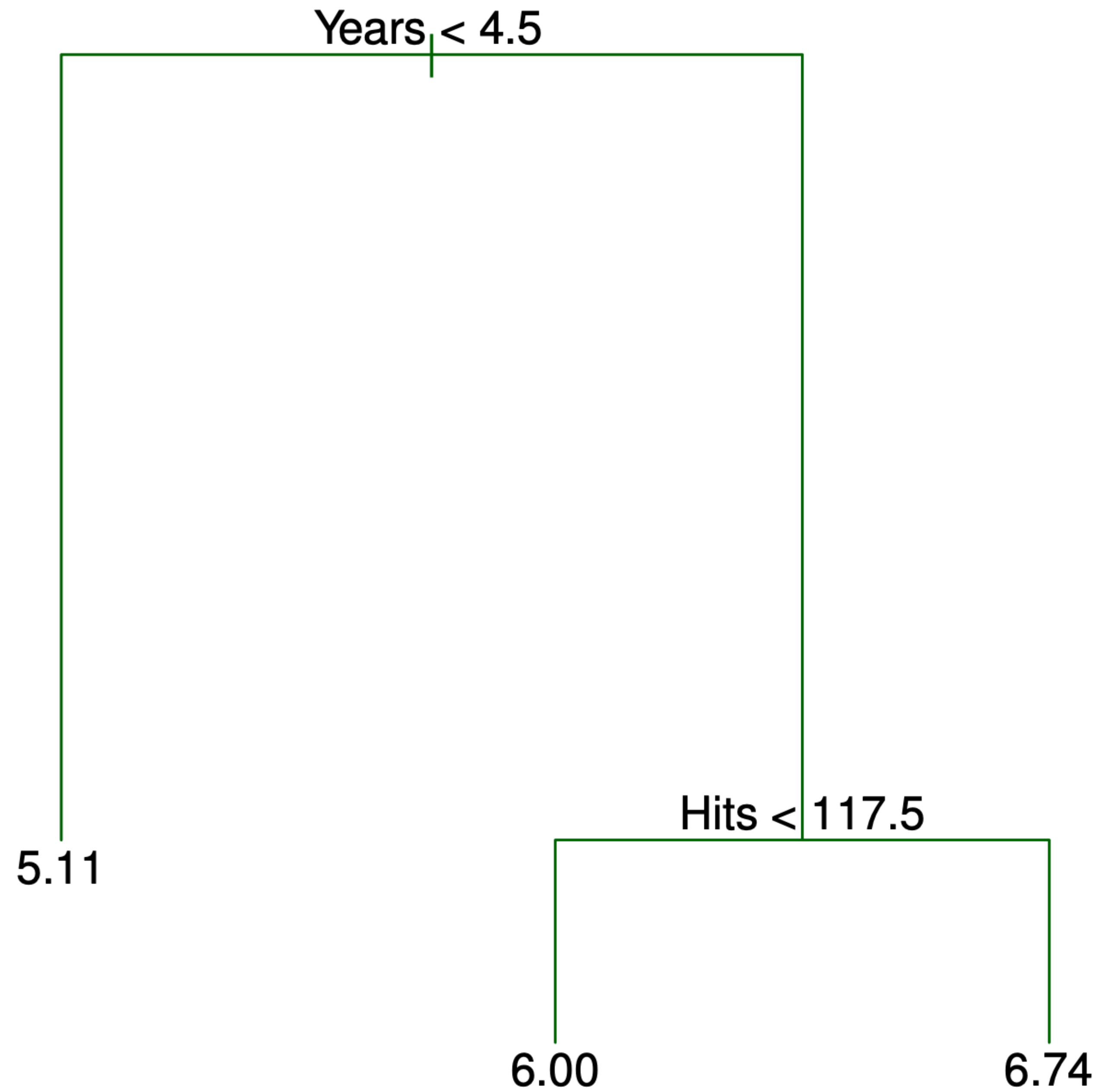
regression trees

procedure

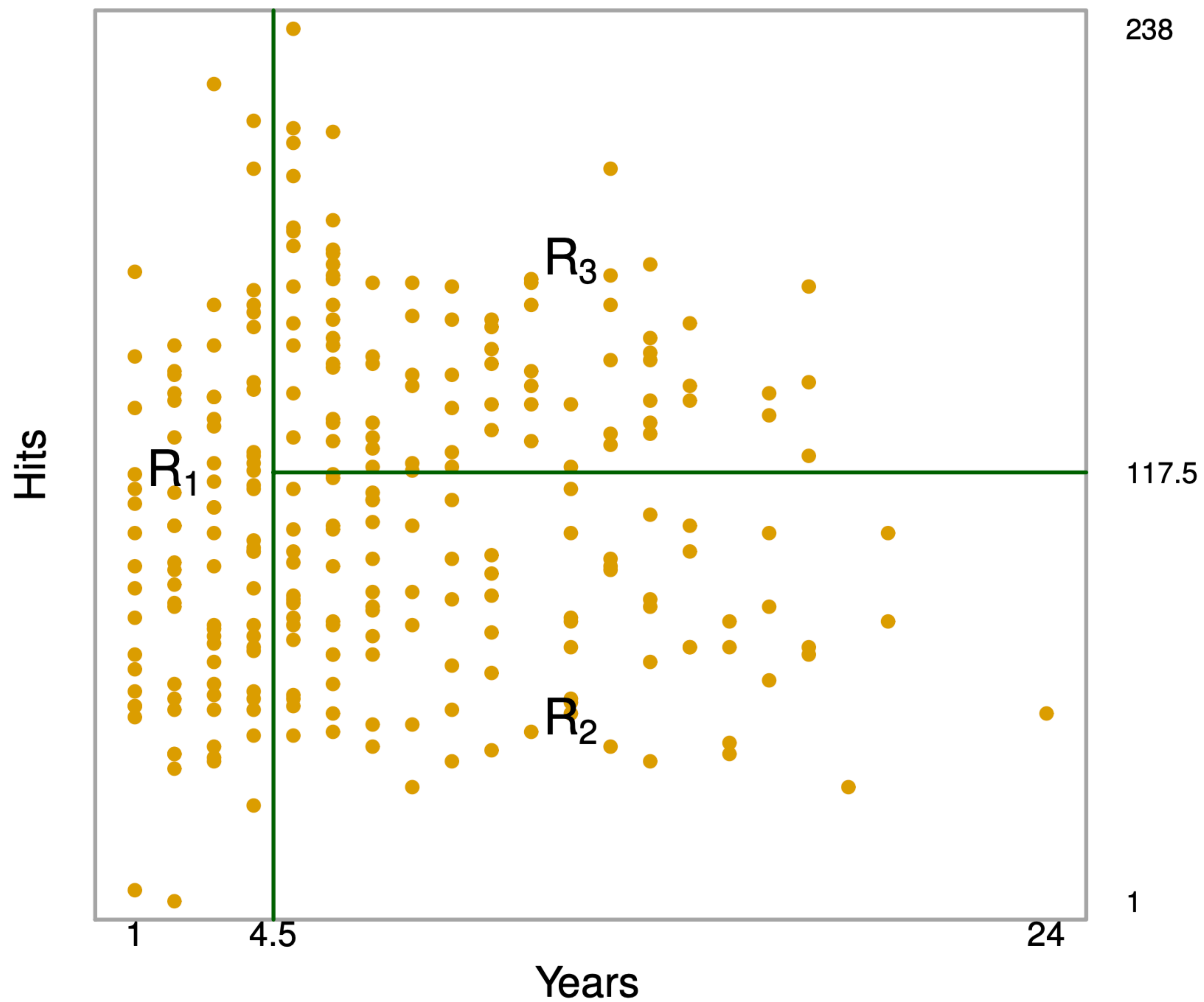
- create root node
- choose a feature that minimises an error metric (e.g., mean squared error)
- create child nodes
- repeat splitting
- assign average of predicted values to leaf nodes

example

- predicting the (log) salary of baseball player
 - number of years played
 - number of hits



- number in leaf node = mean of the corresponding observations



- interpretation of regression tree
 - years is the most important feature
 - players with four years or less and players with 4.5 or more
- among experienced players number of hits becomes a differentiating factor

building the tree

- divide the set of possible feature values ($X_1, X_2, \dots, X_p$) in J distinct non-overlapping regions

minimize $\sum_{j=1}^J \sum_{l \in R_j} (y_l - \hat{y}_{R_j})^2$

$\hat{y}_{R_j}$ is the mean response for the training observations within the j th box

recursive binary split

- for X_j , choose cut point s that reduces sum of squares

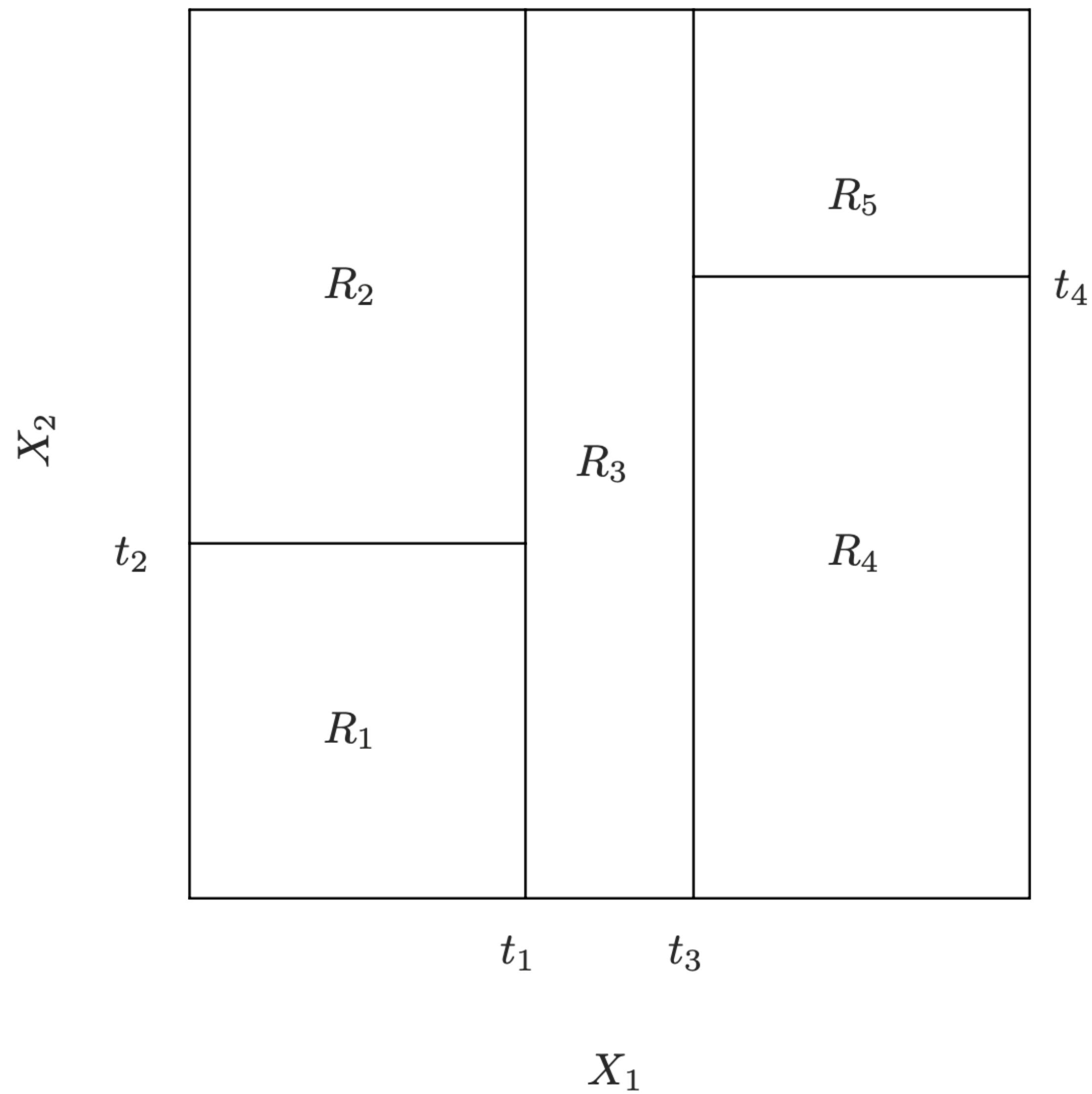
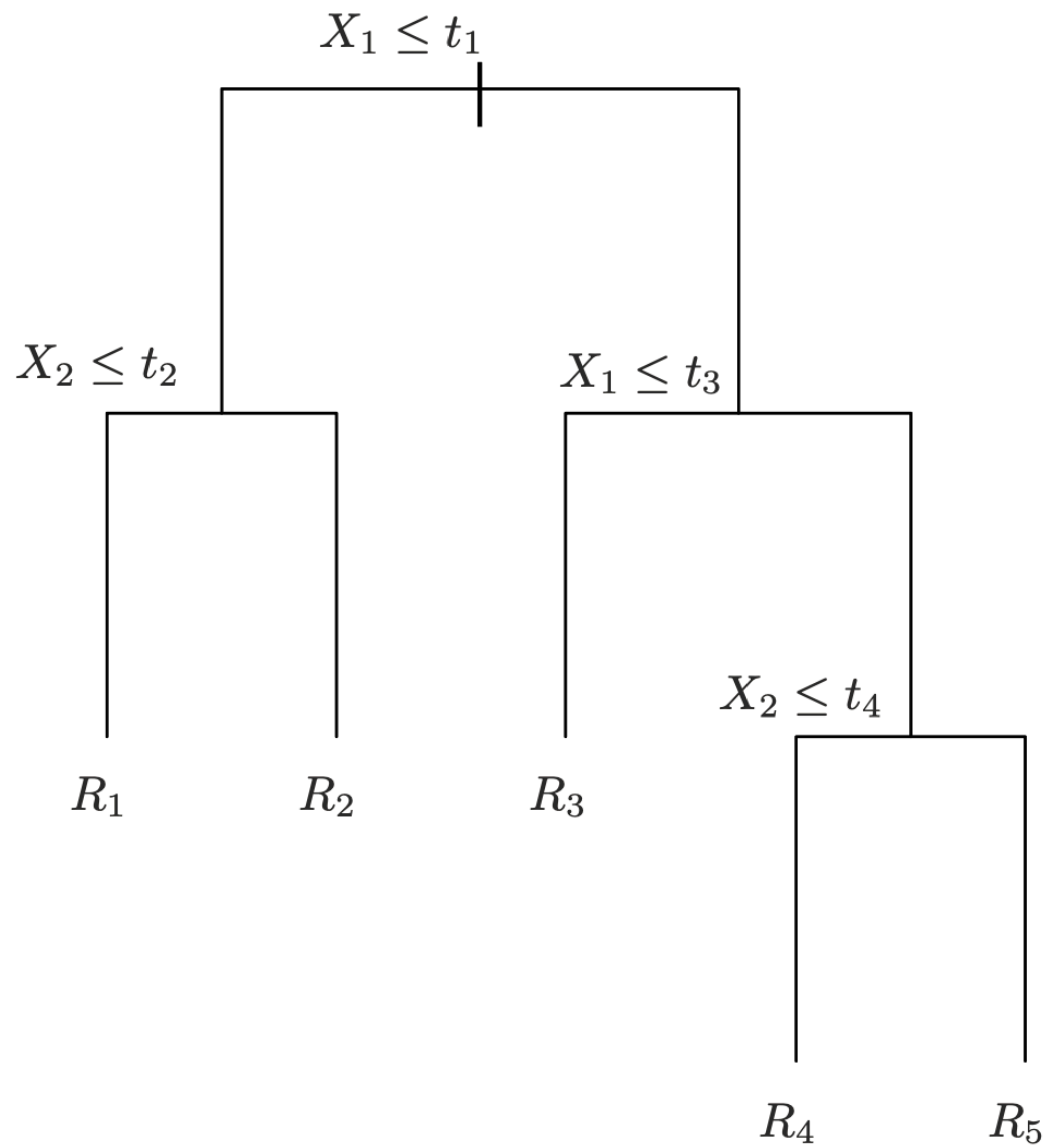
$$R_1 = \{X \mid X_j < s\}$$

$$R_2 = \{X \mid X_j \geq s\}$$

find j and s that minimise

$$\sum_{l:x_l \in R_1(j,s)} (y_l - \hat{y}_{R_1})^2 + \sum_{l:x_l \in R_2(j,s)} (y_l - \hat{y}_{R_2})^2$$

- repeat until a stopping criterion is met



- what to do in case of overfitting?

- set max depth
- set min number of examples in a leaf

- finally choose the mean of each region as **prediction**

what about classification trees?

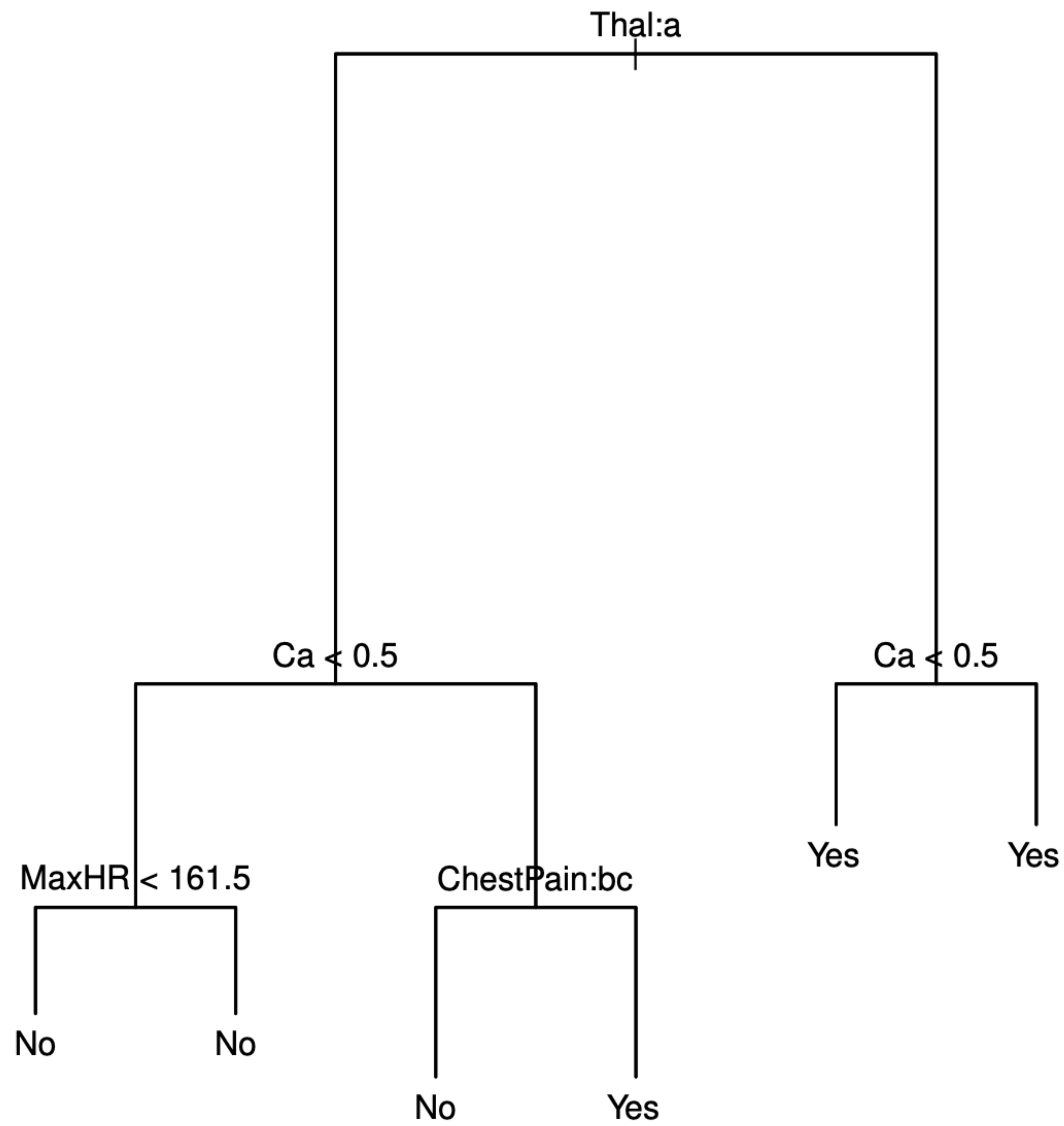
- similar to regression trees
 - but, cannot use sum of squares
 - rather use
 - classification error rate (fraction of training obs that do not belong to the most common class)
 - gini index (total variance across K classes or measure of node purity)
 - entropy (homogeneity of nodes)

$$G = 1 - \sum_{i=1}^k p_i^2$$

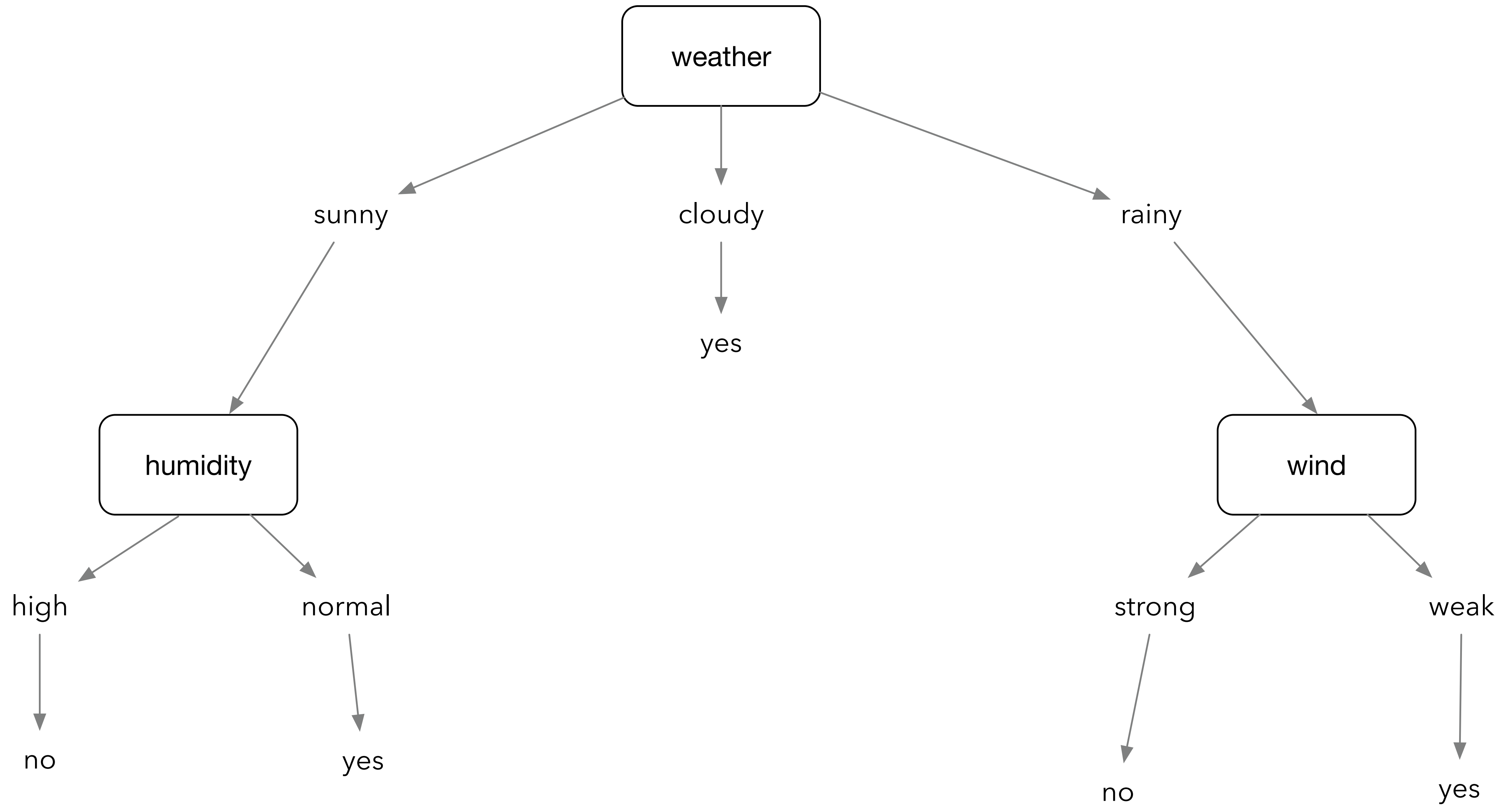
- $G = 0$
 - perfectly pure node
- $G = 0.5$
 - maximum impurity

$$H = - \sum_{i=1}^k p_i \log_2(p_i)$$

$$IG = H(\textit{parent}) - \textit{avg}(H(\textit{children}))$$



Day	Weather	Temperature	Humidity	Wind	Play?
1	sunny	hot	high	weak	no
2	cloudy	hot	high	weak	yes
3	sunny	mild	normal	strong	yes
4	cloudy	mild	high	strong	yes
5	rainy	mild	high	strong	no
6	rainy	cool	normal	stromg	no
7	rainy	mild	high	weak	yes
8	sunny	hot	high	strong	no
9	cloudy	hot	normal	weak	yes
10	rainy	Mild	high	strong	no



limitations

- lack robustness (change in data can completely change the tree)
- less accuracy than other prediction methods
- solution
 - use multiple trees

ensemble methods

- combine simple models into a more powerful one

bagging

regression

- generate B different bootstrapped training datasets
- train the method on the b th bootstrapped dataset and get the prediction
- average all predictions

$$\hat{f}_{bag}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^{*b}(x)$$

- What about classification trees?

- take the majority vote (the most commonly occurring class)

random forests

- improvement over bagging
 - uses a random selection of m predictors (features) during a split

boosting

- sequentially improve the model by updating the residuals
 - the next model puts emphasis on incorrectly predicted observations in the previous model
- adaptive boosting (adaboost; increase weights of incorrect predictions)
- gradient boosting (optimise the loss function)
- extreme gradient boosting (xgboost; improve performance)

stacking

- select base models
- train all base models with same training set
- choose meta learner
 - uses predictions from base models to make final prediction

in short...

- decision trees
 - Regression
 - classification
- ensemble methods
 - bagging
 - boosting
 - stacking